

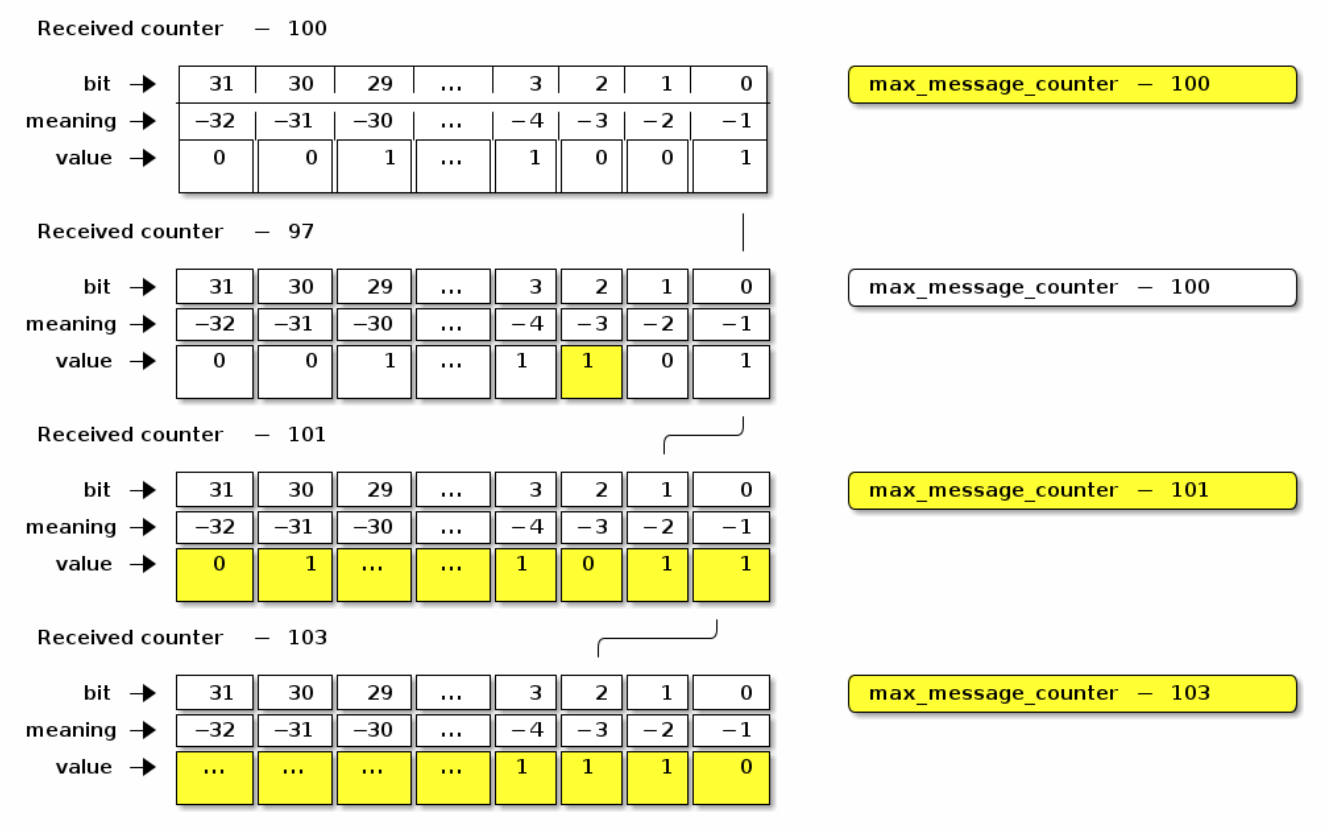


Figure 7. Message Reception State Example

4.6.5.2. Use of Message Reception State for Encrypted Messages

The algorithm for querying and updating message reception state varies slightly depending on whether the system is tracking reception of encrypted messages or unencrypted messages.

Message Counters with maximum

For encrypted messages of [Secure Unicast Session Type](#), any arriving message with a counter in the range $[(\text{max_message_counter} + 1) \text{ to } (2^{32} - 1)]$ SHALL be considered new, and cause the `max_message_counter` value to be updated. Message counters within the range of the bitmap SHALL be considered duplicate if the corresponding bit offset is set to true. All other message counters SHALL be considered duplicate.

Message Counters with rollover

A message counter with rollover is a free running message counter that monotonically increases, but rolls over to zero when it exceeds the maximum value of the counter (32-bits). Group keys are secured by a shared, global message counter with rollover as described in [Section 4.6.1.3, “Global Group Encrypted Message Counters”](#).

For encrypted messages of [Group Session Type](#), any arriving message with a counter in the range $[(\text{max_message_counter} + 1) \text{ to } (\text{max_message_counter} + 2^{31} - 1)] \pmod{2^{32}}$ SHALL be considered new, and cause the `max_message_counter` value to be updated. Messages with counters from $[(\text{max_message_counter} - 2^{31}) \text{ to } (\text{max_message_counter} - \text{MSG_COUNTER_WINDOW_SIZE} - 1)] \pmod{2^{32}}$ SHALL be considered duplicate. A message counter equal to `max_message_counter` SHALL be considered duplicate. Message counters within the range of the bitmap SHALL be considered duplicate